Spatial-Spectral Residuals Informed Diffusion Neural Operator for Pan-sharpening

Jiahan Huang[1,†], Ran Ran[2,†], Junming Hou[1,†,‡], Zihao Chen[1],
Xiaofeng Cong[1], Junling Li[1,‡], Liang-Jian Deng[2,3]
[1]Southeast University [2]University of Electronic Science and Technology of China
[3] Multi-Hazard Early Warning Key Laboratory of Sichuan Province
jiahan_huang@seu.edu.cn, ranran@std.uestc.edu.cn, junming_hou@seu.edu.cn, zihaochen@seu.edu.cn,
cxf_svip@163.com, junlingli@seu.edu.cn, liangjian.deng@std.uestc.edu.cn

Abstract

Pan-sharpening, a fundamental image preprocessing technique in remote sensing, aims to generate spatially and spectrally enriched multispectral imagery by integrating complementary information from texture-rich panchromatic (PAN) images and paired low-resolution multispectral (LRMS) counterparts. Although recent generative diffusion models have achieved impressive fusion quality, these performance gains often come with substantial computational costs, rendering them impractical for resource-constrained scenarios common in remote sensing applications. This work introduces a function-space diffusion model built upon a neural operator architecture that achieves compelling performance with promising efficiency. Specifically, our framework replaces the standard attention-based denoising backbone with a Galerkin-type neural operator, significantly reducing computational complexity while maintaining excellent representational capacity. Furthermore, by explicitly integrating pixel-wise spatial-spectral consistency residuals into each reverse diffusion step, our method establishes a fine-grained, closed-loop guidance mechanism that dynamically calibrates spatial details and spectral fidelity throughout the generation process. Extensive experiments on multiple benchmark datasets demonstrate the effectiveness of our approach over state-of-the-art methods.

1. Introduction

High-resolution multispectral (HRMS) imagery has wide-ranging applications in civilian and military fields, including precision agriculture, environmental monitoring, cartography, and reconnaissance [26, 30]. However, it is challenging for existing multispectral satellites to directly capture HRMS images, because of the physical limitations in sensor imaging systems. A common alternative is to acquire a pair of low-resolution multispectral (LRMS) and high-resolution panchromatic (PAN) images of the same scene, each providing complementary spectral and spatial information, respectively. In this context, pan-sharpening technology has been developed to fuse the complementary information from these two modalities to synthesize HRMS images for downstream remote sensing applications, thereby effectively circumventing the hardware constraints.

The last decades have witnessed substantial methodological progress in the field of pan-sharpening, transitioning from traditional model-driven approaches to deep learning paradigms [30]. Classical methods, defined by their fusion principles, include Component Substitution (CS) methods, Multi-Resolution Analysis (MRA) approaches, and Variational Optimization (VO) algorithms. However, they often rely heavily on manually designed priors and unrealistic assumptions, resulting in suboptimal fusion quality that fails to satisfy practical application requirements. In recent years, deep learning (DL)-based approaches, empowered by large-scale datasets, have achieved remarkable progress and significantly outperform traditional algorithms. By leveraging abundant training data, these approaches employ deep neural networks to directly learn complex nonlinear mapping between input PAN/LRMS pairs and their target HRMS counterparts. Consequently, a wide variety of advanced network architectures continue to be developed to improve fusion performance. Among state-of-the-art approaches, generative diffusion models have demonstrated impressive performance improvements, surpassing conventional DL methods. Nevertheless, such advances are accompanied by considerable computational overhead, hindering their practical utility, particularly for hardware-constrained remote sensing satellites. To address this, many efforts have been made to optimize the efficiency of diffusion mod-

† Equal contribution; ‡ Corresponding authors; The code is available.

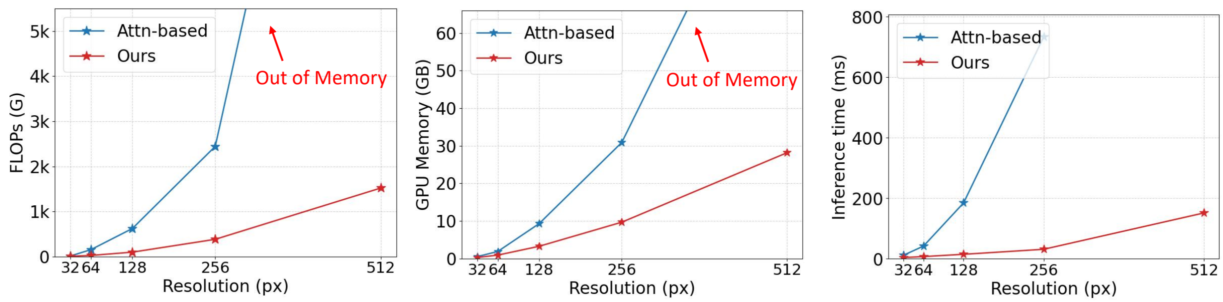

Figure 1. Comparison of FLOPs, Memory Usage, and Inference Time between our neural operator-based denoising framework and the conventional attention-based counterpart [25]. Our approach significantly reduces both FLOPs and memory usage, particularly at larger scales where the attention-based architecture encounters out-of-memory issues. During inference, we achieve a several-fold speedup.

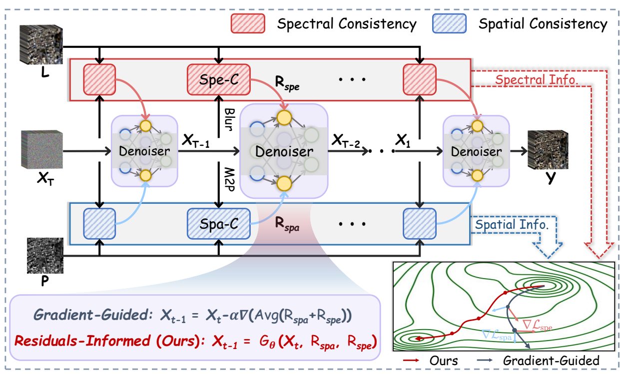

Figure 2. Schematic illustration of the previous gradient-guided strategy and our spatial-spectral residuals informed paradigm. The former uses the gradients of unsupervised losses to update the noise prediction, risking gradient conflicts and requiring tedious weight tuning. Instead, our approach directly incorporates the element-wise spatial-spectral residuals as additional inputs to the denoising network, enabling directed generation of texture-rich and spectrally realistic products.

els, such as low-rank fine-tuning [41] and knowledge distillation [17]. Despite their efficacy, these paradigms still suffer from computational bottlenecks inherent in the standard diffusion architecture: (1) pretraining of self-attention-based diffusion models requires enormous computational resources, and (2) the resulting models remain impractical for real-world deployment, as their substantial storage and inference-time computational demands are incompatible with stringent constraints of satellite hardware.

In this work, we introduce a novel function-space diffusion model built upon a neural operator architecture, to develop an efficient yet effective generative diffusion framework tailored for hardware-constrained pan-sharpening scenarios. As shown in Figure 1, this NO-based diffusion model achieves competitive computational and inference efficiency compared to conventional attention-based architectures. While several recent studies have explored integrating diffusion models with neural operators (NOs) for image super-resolution, they remain largely anchored in the conventional diffusion paradigm. For instance, in [37], the NO serves as an up-sampling operation to provide conditional guidance for diffusion model, and DiffFNO [21] utilizes the NO as a super-resolution backbone further refined by diffusion process. In contrast, our method diverges fundamentally in both motivation and implementation. By situating the diffusion process within the operator learning space, our framework unifies representation learning and generative refinement in a continuous function domain. Specifically, our method follows a two-stage training pipeline. First, we pretrain a discretization-agnostic denoising model using a neural operator architecture conditioned on high-resolution reference images. Subsequently, we fine-tune this pretrained diffusion model using observed PAN/LRMS pairs, dynamically incorporating spatial–spectral residuals to guide the generation toward texture-rich and spectrally realistic HRMS images. During inference, the model iteratively refines the samples by injecting these residuals, computed between the generated HRMS images and the input PAN/LRMS pairs. This strategy differs from previous gradient-guided approaches, which rely on the gradients of unsupervised loss terms to update noise estimates during reverse sampling. Such methods inevitably suffer from gradient conflicts across multiple loss terms and require careful tuning of guidance weights. Instead, our method directly integrates pixel-wise spatial and spectral consistency residuals as dynamic feedback into the sampling process—explicitly steering generation toward texture-rich and spectrally realistic outcomes. Figure 2 depicts the conceptual comparison between our residuals-informed paradigm and previous gradient-guided strategies, with the experimental results on WorldView-3 examples supporting the superiority of our method.

We summarize the main contributions as follows:

- We propose a novel function-space diffusion model built upon a neural operator architecture, situating the diffusion process within the continuous operator learning space.
- Our approach explicitly integrates the element-wise spa-

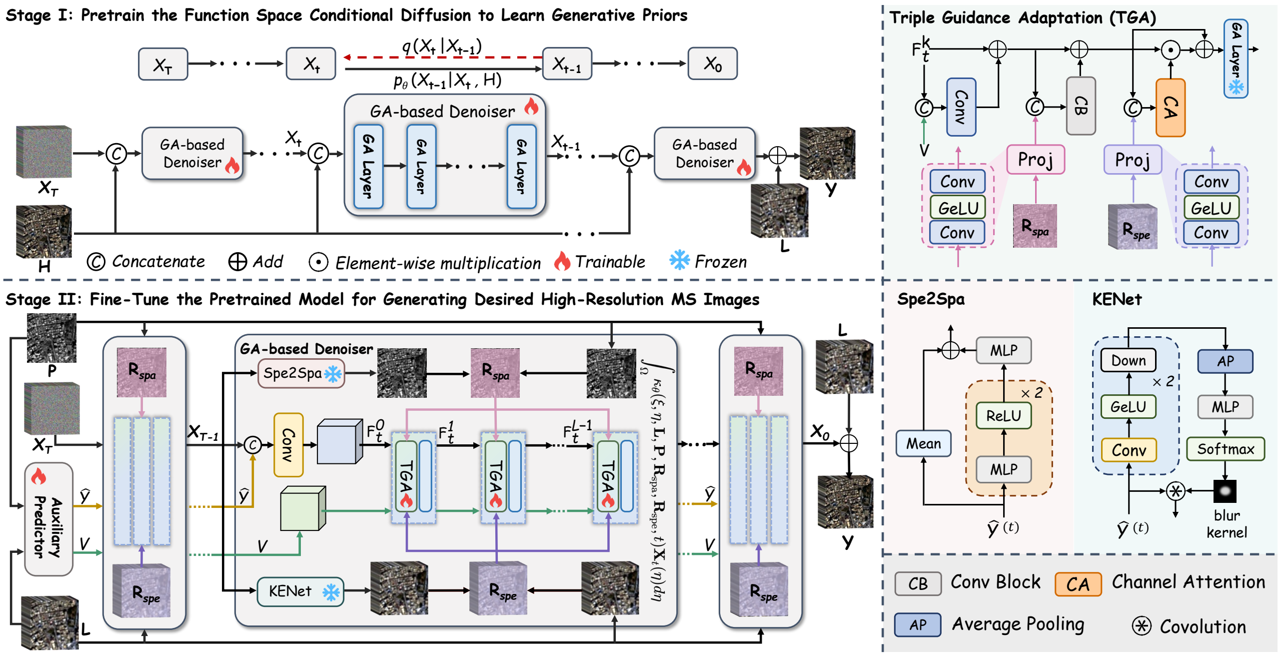

Figure 3. Framework overview of the proposed SRINO, which is implemented through a two-stage training pipeline. Firstly, we pretrain a conditional diffusion model built with Galerkin-type operator layers on high-resolution references to learn high-quality spatial and spectral representations. Subsequently, we introduce a triple guidance adaptation (TGA) strategy, including cross-modality feature $\mathbf{V}$, pixel-wise spatial and spectral consistency residuals, $\mathbf{R}_{\text{spa}}^{(t)}$ and $\mathbf{R}_{\text{spe}}^{(t)}$, to fine-tune the model for generating corresponding HRMS images.

tial and spectral residuals into the denoising network, allowing dynamic refinement of required textures and spectra for high-quality HRMS generation.

- Extensive experiments on multiple benchmark datasets demonstrate that our method outperforms current advanced pan-sharpening approaches, and also delivers a promising performance-efficiency balance compared to conventional attention-based diffusion counterparts.

2. Related Works

Deep Learning-based Pan-sharpening. Fueled by the accessibility of large-scale training data, deep learning has recently dominated pan-sharpening by leveraging its superior ability to model complex nonlinear relationships, achieving substantial performance gains over traditional algorithms [9, 30]. The success of the pioneering PNN model, which delivers competitive performance with a simple architecture, has catalyzed the development of a wide variety of advanced network architectures to further enhance fusion quality, such as convolutional neural networks (CNNs) [7, 14, 38], global transformers [12, 34, 43], and the latest generative diffusion models [5, 13, 17].

Image Super-resolution Neural Operator. Neural operators are newly developed data-driven neural architectures for solving partial differential equations (PDEs). By directly learning mappings between function spaces, they exhibit discretization invariance and resolution-independent generalization [2, 18], rendering them well-suited for tasks involving varying resolutions. Recent years have seen their rapid application in various super-resolution tasks. Pioneered by SRNO [33], many efforts have followed, such as arbitrary-scale scientific data super-resolution [24] and remote sensing pan-sharpening [10, 22]. Recently, several studies have explored integrating neural operators with diffusion models [21, 37]. Despite promising results, they remain within conventional diffusion models. By contrast, we introduce a function-space diffusion model that situates the generative process within a continuous operator learning space using a neural operator denoising architecture.

Diffusion Models for Pan-sharpening. Recently, diffusion models [11, 27] have gained interest in pan-sharpening. Through an iterative denoising process conditioned on multi-source data, such as PAN and LRMS pairs, these methods exhibit superior fusion performance compared to conventional neural networks [5, 17, 25]. However, such improvements often come at the cost of high computational and memory demands, making them impractical for hardware-constrained remote sensing satellites. Furthermore, these methods operate under static conditional information, failing to adapt to evolving needs during the denoising process. Although recent guided diffusion paradigms have been introduced for dynamic guidance [19, 20], they inevitably risk gradient conflicts among multiple loss terms and rely on empirical fine-tuning of the guidance weights. To address these challenges, we introduce an efficient yet effective function-space diffusion model using NO-based denoising architecture, which explicitly integrates element-wise spatial and spectral residuals, steering directed generation toward texture-rich and spectrally realistic outcomes.

3. Methodology

For the sake of simplicity, we first define some notations used in this paper. Let $\mathbf{L} \in \mathbb{R}^{H\times W\times s}$ and $\mathbf{P} \in \mathbb{R}^{H\times W\times 1}$ denote the up-sampled LRMS and PAN images, respectively. $\mathbf{H} \in \mathbb{R}^{H\times W\times s}$ and $\mathbf{Y} \in \mathbb{R}^{H\times W\times s}$ represent the ground truth (originally observed MS image) and fused HRMS images. Here, H and W are the spatial sizes of the image, s is the spectral channel number of MS.

3.1. Overview

As illustrated in Fig. 3, our framework follows a two-stage training pipeline [39] comprising (1) function-space conditional diffusion and (2) triple guidance adaptation. In the first stage, we pretrain an NO-based conditional diffusion model on high-resolution reference data. The goal of this stage is to learn a generative prior within a continuous function space, capturing high-quality spatial and spectral representations. The key insight behind this is to steer the diffusion process toward a spatially and spectrally reliable solution space, thereby establishing a robust foundation for the subsequent pan-sharpening tasks. In the second stage, we freeze the parameters of the pretrained NO-based diffusion model and fine-tune adaptation modules using cross-modality conditional features, along with element-wise spatial-spectral residuals at each reverse sampling step. Crucially, these residuals act as dynamic conditioning cues, enabling the model to perceive and emphasize the required spatial-spectral information during generation. Unlike previous guided diffusion paradigms that rely on external loss functions, our method internalizes the spatial-spectral consistency requirement by architectural design, allowing for dynamic refinement of both spatial details and spectral characteristics throughout the denoising trajectory.

3.2. Function Space Conditional Diffusion

Preliminary of Neural Operator. In neural operator learning, the objective is to learn a mapping $\mathcal{G} : \mathcal{A} \to \mathcal{U}$ between two infinite-dimensional function spaces $\mathcal{A}$ and $\mathcal{U}$, using a finite collection of observed input-output pairs. In practice, the operator mapping $\mathcal{G}$ is approximated by a parameterized surrogate model $\mathcal{G}_\theta$, typically constructed by stacking neural operator layers:

$$\mathcal{G}_\theta = \mathrm{P}_{\text{out}} \circ \mathrm{M}_L \circ ... \circ \mathrm{M}_2 \circ \mathrm{M}_1 \circ \mathrm{P}_{\text{in}}, \tag{1}$$

where P_{in} and P_{out} represent element-wise projection layers that map between the input/output functions and latent representations, and M_l denotes the l-th operator layer. The general form of each layer is defined as:

$$\phi_{l+1}(\xi) = \phi_l(\xi) + \mathcal{O}\left(\mathcal{K}_l(\phi_l)(\xi) + \phi_l(\xi)\right), \tag{2}$$

where $\mathcal{K}_l$ denotes a kernel integral operator modeling global interactions, and $\mathcal{O}$ is point-wise operator implemented as a feed-forward network. The kernel integral operator forms the theoretical core, defined continuously as:

$$\mathcal{K}(\phi)(\xi) = \int_\Omega \kappa(\phi(\xi), \phi(\eta))\phi(\eta)d\eta, \tag{3}$$

which can be discretized as:

$$\mathcal{K}(\phi)(\xi) \approx \sum_{i=1}^{N} \kappa(\phi(\xi), \phi(\eta_i))\phi(\eta_i), \quad \forall \xi \in \Omega_{h_f}. \tag{4}$$

Note that Ω denotes the continuous spatial domain, and $\xi, \eta \in \Omega$ are continuous coordinates. The discretization in (4) evaluates the integral on a high-resolution grid Ω_{h_f}, where $\eta_i \in \Omega_{h_f}$. Here, $N = H \times W$ denotes the number of pixels on the high-resolution grid.

By parameterizing the kernel κ as a query–key–value inner product, this formulation naturally leads to an attention mechanism, which can be defined as:

$$\kappa(\phi(\xi), \phi(\eta_i)) = \frac{\exp\left(\frac{\langle W_q\phi(\xi), W_k\phi(\eta_i)\rangle}{\sqrt{d_v}}\right)}{\sum_{k=1}^{N}\exp\left(\frac{\langle W_q\phi(\eta_k), W_k\phi(\eta_i)\rangle}{\sqrt{d_v}}\right)} W_v, \tag{5}$$

where W_q, W_k, and W_v are learnable projection matrices. For computational efficiency especially in high-resolution settings, we adopt Galerkin-type attention [4], which approximates the output as:

$$\phi_{\text{out}} = Q(\tilde{K}^\top \tilde{V})/N, \tag{6}$$

with $Q = W_q\phi$, $\tilde{K} = \mathrm{Norm}(W_k\phi)$, $\tilde{V} = \mathrm{Norm}(W_v\phi)$. This linear attention variant significantly reduces complexity from $\mathcal{O}(N^2)$ to $\mathcal{O}(Nd_v^2)$ while maintaining global receptive fields, enabling efficient learning of continuous operator mappings.

Training NO-based Conditional Diffusion Model. We implement the denoising network using a Neural Operator $\mathcal{G}_\theta$ constructed with cascaded Galerkin-attention layers. This architecture operates in a continuous function space, effectively capturing high-quality spatial and spectral generative priors essential for pan-sharpening. Following established methodology [5, 32], we define the target as the residual between the HRMS image and the LRMS image:

$$\mathbf{X}_0 = \mathbf{H} - \mathbf{L}. \tag{7}$$

In the forward diffusion process, the residual $\mathbf{X}_0$ is progressively perturbed by Gaussian noise over T steps:

$$\begin{aligned} q(\mathbf{X}_t \mid \mathbf{X}_{t-1}) &= \mathcal{N}\big(\sqrt{\alpha_t}\,\mathbf{X}_{t-1},\, (1-\alpha_t)\mathbf{I}\big), \\ \mathbf{X}_t &= \sqrt{\bar{\alpha}_t}\,\mathbf{X}_0 \;+\; \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\varepsilon}, \end{aligned} \tag{8}$$

where $\bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i$ and $\boldsymbol{\varepsilon} \sim \mathcal{N}(0, \mathbf{I})$. The reverse process is parameterized by the NO-based denoiser $\mathcal{G}_\theta$, which directly predicts the residual $\widehat{\mathbf{X}}_0 = \mathcal{G}_\theta(\mathbf{X}_t, \mathbf{H}, t)$. The posterior distribution of the denoising process is written as:

$$
\begin{aligned}
p_\theta(\mathbf{X}_{t-1} \mid \mathbf{X}_t, \mathbf{H}) &= \mathcal{N}\big(\mu_\theta(\mathbf{X}_t, \mathbf{H}, t),\ \sigma_t^2 \mathbf{I}\big), \\
\mu_\theta(\mathbf{X}_t, \mathbf{H}, t) &= \frac{\sqrt{\bar{\alpha}_{t-1}}\,(1-\alpha_t)}{1-\bar{\alpha}_t}\widehat{\mathbf{X}}_0 + \frac{\sqrt{\alpha_t}\,(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\mathbf{X}_t, \\
\widehat{\mathbf{X}}_0 &= \mathcal{G}_\theta(\mathbf{X}_t, \mathbf{H}, t).
\end{aligned} \tag{9}
$$

The model is trained using $\mathbf{H}$ as the conditional input to learn reliable generative priors in the function space. The training objective is defined as:

$$
\mathcal{L}_{\mathrm{I}} = \mathbb{E}_{t, \mathbf{X}_0, \boldsymbol{\varepsilon}} \left\| \mathbf{X}_0 - \mathcal{G}_\theta\Big(\mathbf{X}_t,\ \mathbf{H},\ t\Big) \right\|_1. \tag{10}
$$

After T reverse steps, the final HRMS image is reconstructed by adding the estimated residual to the LRMS input, *i.e.*, $\mathbf{Y} = \widehat{\mathbf{X}}_0 + \mathbf{L}$.

3.3. Triple Guidance Adaptation

The pretraining stage enables the model to encapsulate rich spatial-spectral prior knowledge. To fully leverage this generative prior for pan-sharpening learning, we freeze the parameters of the pretrained NO-based denoiser and fine-tune it through triple conditioning guidance, including detail enhanced cross-modal feature, spatial consistency residual, and spectral consistency residual. As illustrated in Figure 3, we first employ a lightweight auxiliary predictor Φ, composed of several residual blocks, to extract the cross-modal feature. Unlike previous approaches that often design dedicated modules to supply distinct cross-modal conditions to the encoder and decoder of the denoising network (*e.g.*, feeding cross-modal features to the encoder and high-frequency details to the decoder) [5, 17], we streamline the integration of cross-modal information. Specifically, we transform the LRMS image into a pseudo PAN image via a pretrained spectral-to-spatial mapping network $f_{\mathrm{M2P}}(\cdot)$ [3, 6]. Then we compute the residual between the real PAN and the pseudo PAN, which captures texture details absent in the LRMS. This residual is concatenated with the original PAN/LRMS image pairs, and fed into the auxiliary predictor Φ. The process is formally defined as:

$$
\begin{aligned}
\mathbf{V} &= \Phi([\mathbf{P}, \mathbf{L}, (\mathbf{P} - f_{\mathrm{M2P}}(\mathbf{L}))]), \\
\widehat{\mathbf{H}} &= \mathrm{Conv}_{3\times3}(\mathbf{V}) + \mathbf{L},
\end{aligned} \tag{11}
$$

where $[\cdot]$ denotes channel-wise concatenation. $\mathbf{V}$ represents the fused cross-modal feature, and $\widehat{\mathbf{H}}$ is the estimated pseudo HRMS image used to replace $\mathbf{H}$ in Stage I.

At step t, we first concatenate the pseudo HRMS $\widehat{\mathbf{H}}$ with the current noisy input $\mathbf{X}_t$, which are then processed by a convolutional layer to obtain the initial noisy feature $\mathbf{F}_t^0$. Next, we progressively integrate triple conditioning guidance, including cross-modal feature $\mathbf{V}$, spatial consistency residual $\mathbf{R}_{\text{spa}}$, and spectral consistency residual $\mathbf{R}_{\text{spe}}$, into each layer of the frozen NO-based denoiser. Specifically, at the l-th denoising layer (l = 0,1,...,L-1), we first integrate the cross-modal feature with current noisy feature $\mathbf{F}_t^l$ through convolutional layers:

$$
\widetilde{\mathbf{F}}_t^l = \mathrm{Conv}_{3\times3}([\mathbf{F}_t^l, \mathbf{V}]) + \mathbf{F}_t^l, \tag{12}
$$

where $\widetilde{\mathbf{F}}_t^l$ denotes the enhanced noisy feature.

Pixel-wise Spatial Consistency Residual. At reverse step t, the model takes as input the previously predicted clean estimate $\widehat{\mathbf{X}}_0^{(t+1)}$. This clean estimate is added to the LRMS image to reconstruct the HRMS estimate at step $t+1$:

$$
\widehat{\mathbf{Y}}^{(t+1)} = \mathbf{L} + \widehat{\mathbf{X}}_0^{(t+1)}. \tag{13}
$$

Following the unsupervised learning paradigm in pan-sharpening [3, 6], we employ the above spectral-to-spatial mapping network $f_{\mathrm{M2P}}(\cdot)$ to transform the previous HRMS estimate $\widehat{\mathbf{Y}}^{(t+1)}$ into a pseudo PAN image, $\widehat{\mathbf{P}}^{(t+1)}$. The spatial consistency residual is then defined as the difference between the real PAN image and this pseudo estimate:

$$
\mathbf{R}_{\text{spa}}^{(t)} = \mathbf{P} - \widehat{\mathbf{P}}^{(t+1)} = \mathbf{P} - f_{\mathrm{M2P}}(\widehat{\mathbf{Y}}^{(t+1)}). \tag{14}
$$

Here, $\mathbf{R}_{\text{spa}}^{(t)}$ is the spatial consistency residual, which is further projected into the feature space via two 3×3 convolutional layers and a GeLU activation in between. The resulting spatial residual features are then integrated with the previously enhanced noisy feature $\widetilde{\mathbf{F}}_t^l$:

$$
\mathbf{Z}_t^l = \mathrm{CB}(\mathrm{Cat}[\widetilde{\mathbf{F}}_t^l, \mathrm{Proj}(\mathbf{R}_{\text{spa}}^{(t)})]) + \widetilde{\mathbf{F}}_t^l, \tag{15}
$$

where $\mathrm{Proj}(\cdot)$ and $\mathrm{CB}(\cdot)$ represent the function of feature-space projection and the convolutional block, respectively. $\mathbf{Z}_t^l$ is the spatial consistency residual enhanced feature.

Pixel-wise Spectral Consistency Residual. To enforce spectral consistency, we employ a spectral kernel estimation network $f_{\mathrm{KE}}(\cdot)$ [36] to generate an 11×11 blur kernel, which is applied to the HRMS estimate $\widehat{\mathbf{Y}}^{(t+1)}$. The spectral residual is then computed as the difference between the blurred HRMS and the original LRMS image:

$$
\mathbf{R}_{\text{spe}}^{(t)} = (f_{\mathrm{KE}}([\mathbf{P}, \mathbf{L}]) \circledast \widehat{\mathbf{Y}}^{(t+1)}) - \mathbf{L}, \tag{16}
$$

where $\circledast$ denotes the convolution operation, $\mathbf{R}_{\text{spe}}^{(t)}$ is the obtained spectral consistency residual. Similar to the spatial branch, we project $\mathbf{R}_{\text{spe}}^{(t)}$ into the feature space and integrate it with the spatially-enhanced feature $\mathbf{Z}_t^l$. The resulting features are further modulated by channel-wise weights and processed through the frozen denoiser layer M_l to produce

Method	Reduced Resolution				Full Resolution		
	PSNR($\pm$ std)	SAM($\pm$ std)	ERGAS($\pm$ std)	Q2^n($\pm$ std)	D_λ($\pm$ std)	D_s($\pm$ std)	HQNR($\pm$ std)
BDSD-PC [29]	32.969±2.784	5.429±1.823	4.698±1.617	0.829±0.097	0.063±0.024	0.073±0.036	0.870±0.053
BT-H [23]	33.080±2.880	4.920±1.425	4.579±1.496	0.832±0.094	0.057±0.023	0.081±0.037	0.867±0.054
LRTCFPan [35]	33.613±2.839	4.737±1.412	4.315±1.442	0.846±0.091	**0.018±0.007**	0.053±0.026	0.931±0.031
FusionNet [7]	38.042±2.592	3.325±0.698	2.467±0.645	0.904±0.090	0.024±0.009	0.036±0.014	0.941±0.020
LAGConv [16]	38.568±2.789	3.104±0.559	2.300±0.613	0.910±0.091	0.037±0.015	0.042±0.015	0.923±0.025
Fourmer [44]	38.268±2.727	3.236±0.681	2.419±0.665	0.911±0.090	0.022±0.010	0.035±0.004	0.944±0.013
HFIN [28]	38.534±2.786	3.088±0.635	2.306±0.557	0.912±0.089	0.025±0.008	0.043±0.017	0.934±0.024
HOIF [45]	38.352±2.855	3.186±0.643	2.385±0.668	0.913±0.086	0.039±0.021	0.039±0.010	0.924±0.026
PanDiff [25]	37.860±2.773	3.316±0.674	2.492±0.656	0.906±0.088	0.027±0.012	0.054±0.026	0.920±0.036
LFormer [12]	39.075±2.844	2.899±0.584	2.165±0.509	$\underline{0.919\pm0.086}$	0.037±0.022	0.036±0.012	0.929±0.027
ADWM [15]	$\underline{39.170\pm2.878}$	$\underline{2.914\pm0.589}$	$\underline{2.145\pm0.531}$	0.919±0.086	0.024±0.010	**0.029±0.015**	$\underline{0.948\pm0.021}$
SRINO (ours)	**39.305±2.882**	**2.869±0.589**	**2.111±0.524**	**0.922±0.084**	$\underline{0.019\pm0.008}$	$\underline{0.032\pm0.012}$	**0.950±0.017**
Ideal value	**∞**	**0**	**0**	**1**	**0**	**0**	**1**

Table 1. Quantitative results for reduced and full resolution WV3 samples, comparing several representative state-of-the-art methods. Bold: Best; Underline: Second best.

the output feature $\mathbf{F}_t^{l+1}$:

$$\begin{aligned} \widetilde{\mathbf{Z}}_t^l &= \mathrm{Conv}_{1\times1}([\mathbf{Z}_t^l, \mathrm{Proj}(\mathbf{R}_{\text{spe}}^{(t)})]), \\ \omega_t^{l+1} &= \sigma(\mathrm{MLP}([\mathrm{GAP}(\widetilde{\mathbf{Z}}_t^l), \mathrm{GMP}(\widetilde{\mathbf{Z}}_t^l)])), \\ \mathbf{F}_t^{l+1} &= \mathrm{M}_l(\mathbf{Z}_t^l \odot \omega_t^{l+1} + \mathbf{Z}_t^l), \end{aligned} \tag{17}$$

where $\mathrm{Conv}_{1\times1}$ is a point-wise convolutional layer, $\mathrm{GAP}(\cdot)$ and $\mathrm{GMP}(\cdot)$ represent channel-wise global average pooling and global max pooling. $\sigma(\cdot)$ is the sigmoid activation function, $\odot$ indicates element-wise multiplication. The training objective in this stage is defined as:

$$\mathcal{L}_{\mathrm{II}} = \mathbb{E}_{t,\mathbf{X}_0,\boldsymbol{\epsilon}} \left\| \mathbf{X}_0 - \mathcal{G}_\theta\Big(\mathbf{X}_t, \mathbf{P}, \mathbf{L}, \mathbf{R}_{\text{spa}}^{(t)}, \mathbf{R}_{\text{spe}}^{(t)}, t\Big) \right\|_1. \tag{18}$$

4. Experiments

4.1. Experimental Settings

Datasets and Metrics. To evaluate the effectiveness of our method, we conduct experiments on three widely adopted datasets: WorldView-3 (WV3), GaoFen-2 (GF2), and QuickBird (QB). The training datasets are simulated from original satellite imagery according to Wald's protocol. Both training and test data are available at PanCollection [8]. For reduced-resolution results, we employ four well-recognized reference-based metrics: Peak Signal-to-Noise Ratio (PSNR), Spectral Angle Mapper (SAM) [40], Erreur Relative Globale Adimensionnelle de Synthèse (ERGAS) [31], and Q2^n index [42]. The real-world full-resolution performance is assessed by three non-reference indicators: spectral distortion index (D_λ), spatial distortion index (D_s), and High-Quality Fusion Index (HQNR) [1].

Implementation Details. All experiments are implemented in PyTorch and run on an NVIDIA GeForce RTX 4090 GPU. We train the model using the AdamW optimizer with momentum coefficients 0.9 and 0.999, where the initial learning rate is set to 4.0×10^{-5} and is decayed by a factor of 0.5 every 10,000 iterations. The training batch size is set to 32, using 64×64 image patches. The diffusion process adopts a cosine noise schedule with 500 steps, and we employ the 25-step DDIM sampling strategy during inference.

Baselines. We compare SRINO with several state-of-the-art DL-based methods, including FusionNet [7], LAGConv [16], Fourmer [44], HFIN [28], HOIF [45], PanDiff [25], LFormer [12], and ADWM [15]. Three classical algorithms are selected for a comprehensive comparison: BT-H [23], BDSD-PC [29], and LRTCFPan [35].

4.2. Comparison with State-of-the-art Methods

Evaluation on Reduced Resolution Scene. First, under the reduced resolution, we evaluate the fidelity of the fused HRMS with respect to the ground truth. As summarized in Table 1 (left panel) and Table 2, our SRINO achieves the best performance among all competing pan-sharpening methods across all reference-based metrics, indicating a superior balance between spatial detail reconstruction and spectral preservation. Figure 4 presents RGB composites of the fused results together with the corresponding mean absolute error residual maps with respect to the ground truth. Compared with other deep learning baselines, SRINO pro-

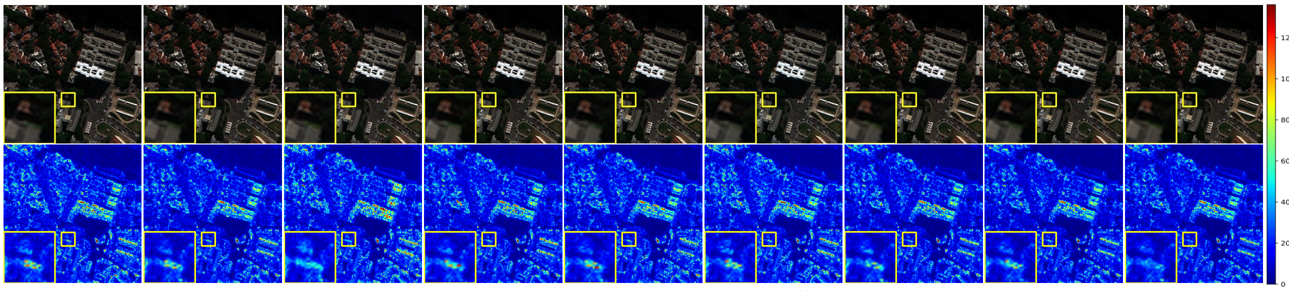

Figure 4. The visual results (top) and mean absolute error maps (bottom) of DL-based methods on a reduced resolution WV3 sample.

Method	Reduced Resolution-GF2				Reduced Resolution-QB			
	PSNR(± std)	SAM(± std)	ERGAS(± std)	$Q2^n$(± std)	PSNR(± std)	SAM(± std)	ERGAS(± std)	$Q2^n$(± std)
BDSD-PC [29]	35.180±2.317	1.681±0.360	1.667±0.445	0.892±0.035	32.547±3.202	8.089±1.980	7.515±0.800	0.831±0.090
BT-H [23]	36.054±2.236	1.649±0.360	1.528±0.409	0.918±0.025	32.648±3.273	7.194±1.552	7.401±0.838	0.833±0.088
LRTCFPan [35]	37.599±2.331	1.298±0.312	1.272±0.343	0.935±0.030	33.260±3.272	7.187±1.711	6.928±0.812	0.855±0.087
FusionNet [7]	39.639±2.270	0.974±0.212	0.988±0.222	0.964±0.009	37.532±2.518	4.923±0.908	4.159±0.321	0.925±0.090
LAGConv [16]	42.735±1.447	0.786±0.148	0.687±0.113	0.980±0.009	38.181±2.456	4.547±0.830	3.826±0.420	0.934±0.088
Fourmer [44]	40.670±1.903	0.976±0.209	0.885±0.185	0.970±0.011	36.797±2.597	5.079±1.009	4.494±0.410	0.924±0.080
HFIN [28]	42.189±1.752	0.843±0.148	0.735±0.126	0.977±0.011	38.247±2.403	4.542±0.805	3.813±0.322	0.934±0.085
HOIF [45]	40.982±1.802	0.943±0.205	0.841±0.162	0.974±0.009	38.242±2.119	4.521±0.811	3.825±0.510	0.933±0.094
PanDiff [25]	42.326±1.635	0.875±0.133	0.727±0.115	0.981±0.008	38.538±2.401	4.524±0.792	3.688±0.343	0.937±0.084
LFormer [12]	<u>44.196±1.800</u>	<u>0.648±0.130</u>	<u>0.578±0.112</u>	<u>0.985±0.007</u>	<u>38.674±2.370</u>	<u>4.382±0.789</u>	<u>3.616±0.314</u>	<u>0.939±0.082</u>
ADWM [15]	43.884±1.714	0.672±0.130	0.597±0.107	<u>0.985±0.006</u>	38.466±2.420	4.450±0.809	3.705±0.346	0.937±0.085
SRINO (ours)	**44.228±1.709**	**0.646±0.129**	**0.573±0.101**	**0.985±0.006**	**38.864±2.346**	**4.351±0.773**	**3.542±0.313**	**0.940±0.085**
Ideal value	∞	**0**	**0**	**1**	∞	**0**	**0**	**1**

Table 2. Quantitative results for reduced resolution GF2 and QB samples, comparing several representative state-of-the-art methods. Bold: Best; Underline: Second best.

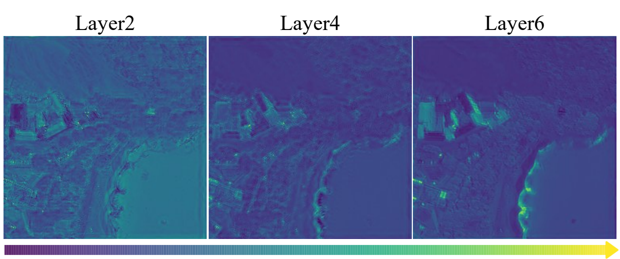

Figure 5. Feature maps across different denoiser layers.

duces sharper structures and fewer spectral artifacts, evidenced by the lowest residual responses in both homogeneous regions and fine-textured areas. As shown in Figure 5, the feature maps progressively gain sharper textures and increasingly distinguishable details as the network advances. These quantitative and qualitative observations confirm the effectiveness of the proposed method.

Evaluation on Full Resolution Scene. We further assess the practical applicability of SRINO on full-resolution data. As shown in the right panel of Table 1, SRINO delivers consistently superior performance across most metrics on the WV3 dataset, mirroring the trends observed in the reduced-resolution setting. Similar behavior is observed on the GF2 and QB datasets, and the corresponding quantitative results are reported in the supplementary material. These full-resolution experiments demonstrate that SRINO generalizes well to real-world pan-sharpening scenarios.

4.3. Ablation Experiments

Ablation on the Spatial and Spectral Residuals. We first analyze the impact of two key components in our model: the spatial consistency residual and spectral consistency residual. To verify their individual and combined contributions, we consider four variants: (i) a baseline without either residual module, (ii) only the spatial residual module enabled, (iii) only the spectral residual module enabled, and (iv) both modules enabled. The quantitative results in Table 3 show that incorporating either module already yields noticeable gains over the baseline, while enabling both spa-

Config.	$\mathbf{R}_{\text{spa}}$	$\mathbf{R}_{\text{spe}}$	PSNR↑	SAM↓	ERGAS↓	$Q2^n$ ↑
I	×	×	39.013	2.945	2.187	0.919
II	✓	×	39.177	2.912	2.145	0.920
III	×	✓	39.124	2.937	2.163	0.920
Ours	✓	✓	**39.305**	**2.869**	**2.111**	**0.922**

Table 3. Ablation study on the core spatial consistency residual $\mathbf{R}_{\text{spa}}$ and spectral consistency residual $\mathbf{R}_{\text{spe}}$ on WV3.

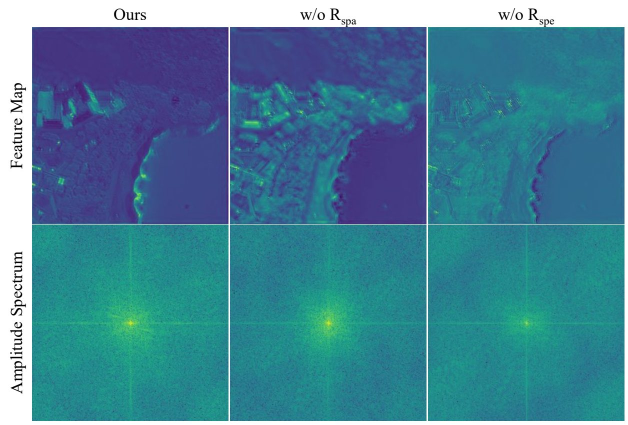

Figure 6. Feature maps and frequency amplitude spectra for the ablation study of the spatial and spectral consistency residuals.

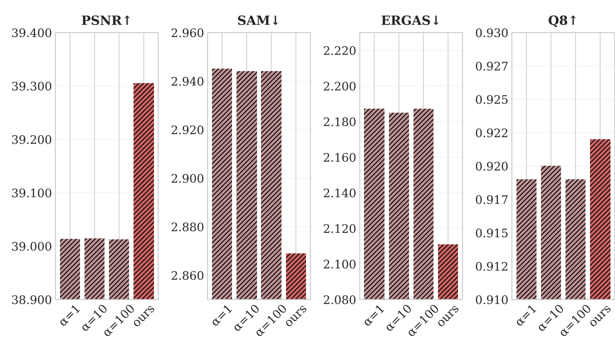

Figure 7. Ablation study on conventional gradient-guided strategy with varying intensities (α=1, α=10, α=100) and our spatial-spectral residual-informed paradigm.

tial and spectral residuals leads to the best performance, confirming that they contribute in a complementary manner. As illustrated in Figure 6, the model incorporating both spatial and spectral consistency residuals produces feature map with clearer structures and rich details compared to those without such residual guidance, as further corroborated by corresponding brighter amplitude spectrum.

Ablation on the Residual Integration Strategy. We compare our residual-informed paradigm with conventional gradient-guided strategy. As shown in Figure 7, our approach yields superior performance over gradient-guided technique. Notably, the latter demonstrates negligible performance variation when changing the guidance weights. In contrast, our approach eliminates gradient computation during inference, mitigating potential gradient conflicts and the need for tedious manual tuning of guidance weights.

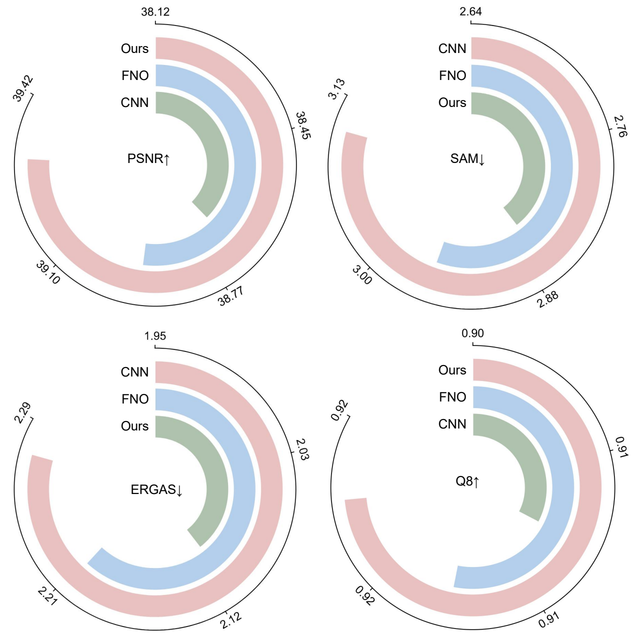

Figure 8. Ablation study on different denoising backbones.

Ablation on Different Denoising Backbones. To assess the impact of the denoising backbone, we replace the proposed Galerkin-type neural operator with a CNN-based backbone and an FNO-based backbone, while keeping all other components fixed. As shown in Figure 8, the CNN variant performs the worst, the FNO variant achieves moderate gains, and ours consistently delivers the best results. This confirms that the proposed backbone is more effective for modeling spatial–spectral dependencies within the diffusion-based pan-sharpening framework.

5. Conclusion

In this work, we propose SRINO, an efficient and effective function space diffusion model tailored for pan-sharpening. Specifically, our model is built upon a neural operator architecture that operates in a continuous function space, achieving promising efficiency and expressiveness suitable for hardware-constrained remote sensing satellites. Furthermore, we directly incorporate spatial-spectral consistency residuals at each denoising step, enabling dynamic refinement of both spatial details and spectral fidelity. Beyond pan-sharpening, this paradigm offers a generalizable solution for broader vision tasks, as it internalizes generative feedback directly into the denoising model, facilitating a more native and consistent generation process.

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